The Limitations of Generative AI

What these systems cannot (yet) do, and why

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Abstract

Generative AI systems now produce fluent text, images, audio, and code at near-zero marginal cost, and public understanding of them is built mostly on demonstrations of success. This survey starts from the opposite end: it catalogues what these systems systematically cannot do, and asks, for each limitation, whether it is fundamental to the design or merely an engineering shortfall awaiting a better fix. The discussion is written for a reader without a machine-learning background, and it keeps one discipline throughout: every claim is separated into what is *observed*, what is *explained*, and what is merely *narrated* — and every citation was re-verified against public bibliographic indexes during preparation, not copied from memory. The taxonomy that emerges is not a flat list but a structure of three kinds of failure. *Statistical* failures — confabulation, missed context, poor calibration — behave like sampling error and respond, at best, to better data and better measurement. *Dynamical* failures arise when a system learns while it acts, so that errors circulate and amplify through feedback instead of averaging out. *Normative* failures are the most distinctive: some obligations on AI systems, such as privacy, are rules rather than probabilities, and a system trained only to imitate text cannot even represent the rule it is breaking. Recognizing which kind of failure one is looking at changes what counts as a remedy — and explains why many announced fixes do not survive contact with the problem they claim to solve. Two further limitations sit outside the models themselves: the dominant delivery mechanism — a hosted service, with weights withheld — has measurable costs for research and for digital sovereignty, and the incentives that selected that architecture shape the claims made about these systems as systematically as they shape how they are built.

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1 Introduction

This document began as an answer to a practical question from someone whose exposure to artificial intelligence is almost entirely through the generative products of the last decade — chat assistants, image generators, code helpers. That exposure is real but asymmetric: the successes are easy to see, and the structure of the failures is not. Someone who has only seen a system succeed cannot tell a reliable tool from an unreliable one, cannot judge announced fixes, and cannot assess the more breathless claims made in both directions. The purpose here is to supply the missing half of the picture.

The scope is deliberately bounded. The primary object is the large language model — the architecture behind text generation — with image, audio, and video generators discussed where they share the same limitations. Where a limitation spills over the border of "generative AI" into machine learning generally, or into questions of governance and law, the text says so rather than pretending the border held.

The method is the part the reader should trust most. Each section states the limitation in plain language, gives the strongest observed evidence for it, and distinguishes three levels of claim: what is directly *observed* (measured in published experiments), what is *explained* (proved or well-argued mechanistically), and what is merely *narrated* (repeated so often it sounds established, but is not). The document also looks outside the models. Section 5.2 examines the delivery mechanism — models offered only as hosted services, weights withheld — and traces both its research costs and its political ones, through the surveillance incidents and the digital-sovereignty programs that have answered them; Section 5.3 asks how the commercial incentives that selected that architecture shape what is claimed about these systems, and finds the misrepresentation question half-measured and half-open.

The bibliography was not assembled from memory: every entry was checked against public indexes (DBLP, the arXiv API, Crossref) by a verification script whose raw output ships alongside this document, and open-access copies of the cited papers are archived with cryptographic hashes. The method, and the errors verification caught, are described in Appendix A. Where the literature disagrees, the disagreement is reported rather than resolved by fiat.

2 Background and terminology

The dominant architecture for language generation is the transformer (Vaswani et al. 2017), trained on a simple objective: given a sequence of text, predict the next piece of text. Repeat that objective at enormous scale over a large fraction of the written internet, and the result is a system that produces startlingly fluent continuations. Text generators of this kind are called *autoregressive*: each output token is produced conditioned on everything so far, one after another, like a very well-informed typist. Image, audio, and increasingly video generators use a different paradigm called *diffusion*: the system learns to turn pure noise into a coherent sample step by step, like a photograph developing — and, as Section 4 discusses, the same idea now works for text.

A few terms recur throughout and deserve fixed meanings.

Hallucination or *confabulation* is the production of statements unsupported by the training data, the conversation, or the world. This document prefers *confabulation* where it matters: the term “hallucination” misleadingly suggests a perception problem, when the mechanism is the training objective itself rewarding plausible-sounding continuation.

Grounding is the connection between language and what it is about. A system trained purely on text has seen descriptions of the world, not the world; the same strings are compatible with many different worlds (Bender and Koller 2020).

Calibration is the match between a system’s expressed confidence and its actual reliability. A well-calibrated system that is right 80% of the time says so; a badly calibrated one sounds equally sure either way.

Benchmark contamination is the leakage of test questions into training data, which turns examinations into memory tests.

Throughout, *limitation* means a systematic constraint on reliability, grounding, or reasoning — not a one-off defect of a particular product, and not an inability that better prompting happens to route around in one lucky case.

3 A taxonomy of limitations

The limitations below are usually presented as a laundry list, which invites the mistaken conclusion that one big fix — more data, more compute, better guardrails — will eventually clear them all. Grouping them instead by *mechanism*, meaning what kind of error they are and how they respond to effort, yields three families. The grouping is the single most useful thing this document offers, and it is summarized in Table 1.

Table 1: The three families of failure and the remedies that actually match them. Offering a statistical remedy for a normative failure is a category error — the most consequential one in current AI policy debates.

Family	**Signature**	**Matching remedy**
Statistical	errors average out over samples	better data, better measurement, better calibration
Dynamical	errors circulate and amplify through feedback	architectural separation of learning from acting
Normative	errors are systematic because the violated rule cannot even be represented	inspectable, typed structure outside the trained weights

Sections 3.1 through 3.8 belong to the statistical family: their errors are, at bottom, sampling errors, and additional data or measurement can in principle dilute them. Section 3.9 is the dynamical family in miniature: there the errors do not average out, because the learner’s own actions shape the data it learns from. The normative family is not a defect *inside* these systems but a structural absence *around* them — there is nothing in a trained model that can represent “this transmission violates this norm” — and it is treated in Section 5.1, after the design space in Section 4 has been laid out.

3.1 Reliability: confabulation and hallucination

Generative models produce statements unsupported by their inputs, their training data, or the world (Ji et al. 2023; Mallen et al. 2023). The phenomenon is not rare noise from a badly built model: it follows from the training objective, which rewards plausible continuations, and plausibility is a matter of what usually sounds right, not what is true. A model asked about a niche subject has a choice between fluent invention and an awkward admission of ignorance, and everything in its training has rewarded the former.

Two observed properties are worth knowing. First, the rate of confabulation depends on the subject: on popular topics, where the training data is dense and largely consistent, models do much better than on the long tail of rare ones, where retrieval of genuine memorized facts collapses (Mallen et al. 2023). Second, fluency is uncorrelated with accuracy — the false statements arrive in exactly the same confident prose as the true ones, which is why readers' natural heuristic of "it sounds like it knows what it is talking about" fails precisely here.

3.2 Reasoning, compositionality, and planning

Transformers degrade sharply on tasks whose solution requires reliable composition of sub-computations (Dziri et al. 2023): if each step is 95% reliable and the answer needs forty steps in sequence, the whole is unreliable in a way that more data on the *whole* does not repair, since the model's errors compound with depth of reasoning rather than averaging out per step. In controlled studies, large language models demonstrably fail classical planning benchmarks — producing invalid plans, plans that fail their own preconditions, or plans that merely paraphrase the ones seen in training (Valmeekam et al. 2023; Kambhampati et al. 2024).

The plain-language version: these systems are superb at *recognizing* what a plan or proof or derivation looks like, and unreliable at *producing* one under constraints. That distinction — pattern recognition versus constrained generation — is the load-bearing one in much of what follows.

3.3 Grounding: form versus meaning

Language-only training leaves meaning underdetermined: the same strings are compatible with many worlds (Bender and Koller 2020). A system can manipulate the word "coffee" flawlessly — translate it, rhyme with it, write sonnets about it — with nothing that connects the word to the substance. Multimodal training (attaching images, audio, or video to text) narrows this gap but does not close it: the association between a caption and a photograph is still a statistical co-occurrence, not a grasp of what makes the photograph a photograph of anything.

For the reader evaluating a product demo, the practical test is to ask questions whose answers cannot be inferred from text alone — how heavy, in which room, what happened *after* the training cutoff — and to notice how confidently the answers arrive anyway.

3.4 Data-derived limits

Everything a generative system "knows" is a compression of its training corpus, so defects in the corpus propagate directly into behavior. The consequences are several and different in kind. *Bias*: systematic skews in the corpus (who writes, who is written about, in what tone) become systematic skews in output (Weidinger et al. 2021). *Temporal cutoff*: a model's world ends at its last data; it will speak confidently of a past that simply stopped. *Duplication and memorization*: heavily repeated text is retained nearly verbatim, which matters both for accuracy and for leaking what should have stayed private. *Long-tail thinness*: rare facts, names, and events are exactly where confabulation flourishes, per Section 3.1. The foundation-model literature treats these as a family because they share a cause — the data is the world as seen from a very partial vantage point (Bommasani, Hudson, et al. 2021).

3.5 Calibration and uncertainty

Models often lack well-calibrated confidence over their own outputs; fluent delivery is a poor proxy for posterior probability (Kadavath et al. 2022). The nuance matters: large models are *partially* calibrated — on some question types their internal states carry usable signals about their own correctness, and models can be trained to surface those signals. But the default interface, a confident prose stream, discards that signal. The user sees the same tone whether the model is reciting a famous theorem or inventing a citation. Until the interface exposes uncertainty honestly — and few do — every downstream consumer of the output inherits an unmeasured risk.

3.6 Evaluation pathologies

How do we know any of the capability claims? Mostly from benchmarks — and benchmarks have documented pathologies of their own. Models learn heuristic shortcuts that score well on test sets without the underlying competence (McCoy et al. 2019); test questions leak into training data, turning examinations into memory tests; and Goodhart's law applies with full force — once a measure becomes a target, it ceases to be a good measure. The composite effect, visible in the repeated pattern of benchmark scores rising while careful probes of the same skill stay flat, is that headline metrics are an unreliable guide to capability. The reader should discount any claim of the form "the new model scores X%" unless the evaluation method is described and contamination is addressed.

3.7 Social and systemic risks

The families above are per-interaction failure modes; their social consequences compound. The first compounding is plain scale: harms to persons and institutions arise when individual failures — stereotyping, privacy violation, misinformation, skewed representation — are emitted at the rate and volume of industrial text production (Weidinger et al. 2021). A single confabulated claim is an irritant; a million per hour is an information environment.

The second compounding is structural. When generators are chained into *agentic* pipelines — one system's output becoming another's input with no human between the stages — error classes propagate instead of being caught: the foundation-model literature treats this as a systemic property of the technology rather than an accident of any one deployment (Bommasani, Hudson, et al. 2021). An error rate that is merely annoying in a chat window becomes consequential when the same class of errors silently flows through software, documents, and decisions, and the reproducibility problems documented for the underlying systems (Kapoor and Narayanan 2023) make the propagation hard to audit after the fact.

The third compounding is the least discussed: fluent prose carries an implicit claim to intention. Human readers have a lifetime's training in the dramaturgy of communication (Bender, Gebru, et al. 2021) — when something speaks in confident, well-formed sentences, we extend it the credibility we would extend a competent person, and we feel betrayed or persuaded rather than merely notified of noise. A misfiled document does not generate outrage; a fluent confabulation does. This asymmetry means the social harm of a statistical failure can exceed its information content, and it is one more reason vigilance cannot be the last line of defense: it puts the burden of skepticism on the least-informed party, at the moment of reading, forever.

This subsection is where the statistical family touches the normative one: scale converts individual sampling errors into systemic outcomes, but as Section 5.1 argues, some of those outcomes are then governed by *rules* that the underlying systems cannot represent.

3.8 Memory and context: the read side of state

Contemporary models do not remember: they are functions of their context window, and long-term memory is externalized to engineering scaffolding — files, databases, retrieval systems bolted on around the model. The *write* side — changing what the system knows through experience — is the dual-control problem of Section 3.9. The *read* side is a distinct failure that persists even with frozen weights: models reliably use information at the edges of long contexts but miss the middle (Liu et al. 2024), and models advertising 32K-plus token windows often fail realistic tasks well below their advertised limits (Hsieh et al. 2024). Context is best understood as short-term working memory with unmeasured fidelity. The retrieval scaffolds sold as the fix inherit the same read-reliability problem one layer out: whatever the retriever fails to find, or the model fails to notice mid-context, is silently absent from the answer — with no change in tone to mark the gap.

3.9 Learning in the loop: dynamical failures

The preceding failures are *statistical*: errors average out over samples. A distinct family arises when a network is updated online inside its own control loop, where errors *circulate through feedback*. Combining bootstrapped targets with function approximation and off-policy data diverges — provably, even with linear approximation and exact dynamic-programming updates (Baird 1995; Tsitsiklis and Van Roy 1997; Hasselt et al. 2018). The acting controller additionally corrupts its own data distribution: agents overfit their earliest experiences (Nikishin et al. 2022), shifted distributions overwrite prior learning (French 1999), and prolonged continual training destroys the *capacity to learn* altogether (Dohare et al. 2024). A learned dynamics model invites optimizing against its own flattering extrapolations (Janner et al. 2019). The classical umbrella is Feldbaum's dual control, intractable even for linear-Gaussian systems (Wittenmark 1995); the principled escape is two-timescale separation of estimation from action (Borkar 1997).

The plain-language summary: a system that rewrites itself while it acts is not just a statistical learner with a faster clock — it is a feedback system, and feedback systems have failure modes (oscillation, collapse, lock-in) that no amount of additional data cures. This is why the memory question of Section 3.8 cannot be solved by simply letting the model update itself, and why the governance architecture of Section 5.1 keeps normative state outside the trained weights.

4 Alternatives in the generator design space

Autoregressive models write like a typist, one token after another; diffusion models sculpt the whole answer at once, refining noise into text like a photograph developing. The distinction is not aesthetic. Because the sculptor works all positions simultaneously, it is measurably faster — over a thousand tokens per second on current hardware, with sub-second first tokens (Khanna et al. 2025) — and because the same denoising mechanism natively handles continuous signals, it extends to sound and images without text being the privileged medium. An 8B-parameter diffusion language model trained on a fraction of the data matches an equivalent autoregressive model (Nie et al. 2025), which is evidence that the autoregressive objective is not the only route to competence.

The typist still holds two advantages, and honest reporting requires naming them. First, very long documents are cheaper to serve autoregressively: the model keeps compact notes about what it has already read, so continuing is inexpensive, while the diffusion approach re-examines the whole document at every refinement step. Second, deliberate step-by-step reasoning — the "think for a long time before answering" style behind the strongest current results — is built on long sequential generation, which is the autoregressive shape. Diffusion has a structural self-correction advantage in principle (it can revise an early part of the answer without regenerating the rest; the typist cannot), but that advantage has not yet won the frontier. The verdict: better on speed and modality, demonstrated; better overall, still an open empirical question whose deciding battlegrounds are reasoning-time compute and long-context serving.

5 What the limitations imply

A catalogue is only useful if it changes decisions. The family structure of Section 3 sorts the mitigation landscape — retrieval augmentation, tool use, verification, neuro-symbolic hybrids — into those that address a limitation, those that merely hide it, and those that are category errors (Gao et al. 2023). Retrieval augmentation attacks data-derived limits (recency, long-tail thinness) but inherits the read-side memory problem of Section 3.8: whatever is retrieved poorly is silently absent. Tool use lets a model delegate what it cannot compute, but delegation is only as reliable as the model's judgment about when to delegate. Verification — checking outputs against a formal or empirical oracle — is the strongest statistical remedy, because it converts plausibility into checked correctness, and it is no accident that the flagship neurosymbolic successes below are verification-shaped. None of the statistical remedies touches the dynamical family, which yields only to architectural separation, or the normative family, which is the subject of the next subsection.

5.1 The neurosymbolic case: governance demands structure

Some obligations on AI systems are not statistical questions. Privacy, in its dominant positive theory — Helen Nissenbaum's contextual integrity (Nissenbaum 2009) — is a system of norms over who may transmit what information about whom, to whom, in which context. When that theory was formalized, it became a deontic logic with temporal operators (Barth et al. 2006): the formalization requires recipient beliefs, role and group structure, retention over time, and probabilities of leak paths for risk assessment. Read the required machinery off the formalization and one gets exactly multi-type, multi-agent, multimodal logic with temporality and probabilism — a profile that no trained statistical model possesses, because nothing in a corpus gradient can represent "this transmission violates this norm in this context."

The distinction that matters is between producing *instances* of protective behavior and maintaining *coherence*. A large model fine-tuned on privacy-conscious text will sometimes refuse to repeat someone's address; it will do so unreliably, without an audit trail, without consistency across conversations and time, and with no mechanism for correction short of retraining. Coherence — norm consistency, auditability, reversibility — requires inspectable structure. The semiotic layer supplies what even the formal theory leaves informal: a typed account of what *kind* of sign is transmitted about whom, since an indexical link (this photo of a face *is* this person) is a doxxing vector, an iconic rendering is representational harm, and a symbolic misuse is a norm violation with an identifiable norm object.

The current state of deployment is worth stating precisely, because both hype and dismissal get it wrong. Narrow hybrids — symbolic structure in load-bearing but restricted roles, such as constrained decoding and rule-based guardrails over model output — are deployed commercially. Flagship neurosymbolic systems exist but at research stage: an olympiad-geometry solver pairing a neural language model with a symbolic deduction engine (Trinh et al. 2024), and formal mathematical reasoning at olympiad level (Hubert et al. 2025). What is deployed nowhere is the full governance-demanding profile — normative, multi-agent, multimodal, temporal. The vacuum is the finding. Architecturally, the precedent for how to fill it is the ethical governor (Arkin 2009): normative state held outside the trained weights, with neural perception fast and graded — the same two-timescale separation that Section 3.9 derives from stability, derived here from accountability.

5.2 The shape of the machine shapes the science

One further limitation is not inside the models but around them. The dominant delivery mechanism of the past decade — models offered only as hosted services, weights withheld — was selected under incentives where keeping the service proprietary had commercial value, and the choice has measurable costs for research. Transparency scores for major developers are low, with the worst indicators exactly where a service model hides things: training data, compute, hardware (Bommasani, Klyman, et al. 2023). Published findings silently break when hosted models are updated or retired without documentation (Pozzobon et al. 2023) — an entire literature can become unreplicable by a quiet API change. Access to the necessary scale was already concentrating industrially before the commercial boom (Ahmed and Wahed 2020), and the reproducibility crisis in machine-learning-based science compounds the effect (Kapoor and Narayanan 2023). An internal document leaked from Google in 2023 acknowledged the moat reasoning directly (Patel and SemiAnalysis 2023).

Honesty requires marking the boundary of what the evidence shows. Competing explanations — safety review, legal exposure — predict much the same observable behavior as moat preservation, so the *motive* must remain an open question; the measurable damage to the research commons is the part that can be stated without hedging. The consequences have never stayed hypothetical. A hosted service runs on the operator's infrastructure, under the operator's jurisdiction, and the political history of that arrangement has kept arriving. In 2013 the National Security Agency was found to have monitored the German chancellor's mobile phone from the United States embassy in Berlin (Der Spiegel 2013). In 2020 the Central Intelligence Agency was revealed to have secretly owned Crypto AG, a Swiss vendor whose encryption machines served governments around the world (Miller and Pomfret 2020). In 2021, reporting on a leaked Danish intelligence inquiry showed American partners using Danish cables to spy on European politicians, the German chancellor among them (BBC News 2021). In 2024 a joint investigation by +972 Magazine, Local Call, and the Guardian documented a nine-year campaign by Israeli officials and intelligence services to surveil and intimidate International Criminal Court staff and witnesses in order to derail the court's Palestine investigation (Iraqi et al. 2024). The European Parliament had drawn the corresponding conclusion two decades earlier, recommending encryption when its investigation into the ECHELON interception system confirmed it existed (European Parliament 2001). And the Court of Justice of the European Union has invalidated the legal instruments meant to legitimize transatlantic data flows more than once on exactly this ground — the second time, in the 2020 *Schrems II* judgment, holding that United States surveillance law did not ensure protection essentially equivalent to European law (Court of Justice of the European Union 2020).

The lesson generalizes past any single adversary: the incidents implicate allies and enemies alike, because the exposure is a property of who operates the infrastructure and where it sits, not of who currently holds which alliance. A state that has watched one nominal ally surveil it has, if anything, stronger reason to treat every foreign-operated service — including AI delivered from abroad — as an intelligence surface to be defended against.

Out of this history grew a named policy program: *digitale Souveränität* in German, *souveraineté numérique* in French, digital sovereignty in the European Commission's English. The scholarship traces the concept's migration from activist and community usage — where it named autonomy and self-determination in shared infrastructure — into state-centric policy discourse (Couture and Toupin 2019), and the European version is explicitly framed as reducing dependence on foreign, chiefly American, infrastructure (Pohle and Thiel 2020). Its practice is visible in procurement: the German state of Schleswig-Holstein is replacing Microsoft products across its administration with Linux and LibreOffice (European Commission, Interoperable Europe Portal 2026), Denmark's digital agency has announced plans to replace Microsoft products with open-source alternatives for the same stated reasons (The Record 2025), and the EU's Data Act obliges vendors to make switching providers and exporting data practical rather than nominal

(European Parliament and Council of the European Union 2023).

Contemporary generative AI arrived into exactly this weather. It is consumed almost entirely as a hosted service, offered by a small number of predominantly American firms — and the same disclosures that started the sovereignty turn are the reason European policymakers treat that presentation as a dependency question rather than a convenience. Read this way, the research costs documented above and the political costs are two faces of one architecture: the same opacity that makes a service hard to study makes it hard to audit domestically, and the same withheld weights that make findings unreplicable make the deployed function something the customer does not possess.

Independently of geopolitics, the service model changes what buying anything means. A self-contained product keeps working when its maker does not; a service that shuts down — for the ordinary commercial reasons providers shut down: acquisition, repricing, lost interest, bankruptcy — takes its function with it, and the customer's data and workflow go with both. This is not distinctive to AI; it is the standard lifecycle risk of hosting, which is precisely why the sovereignty programs above legislate switching and portability. But regulation reacts to a market shape that was already chosen, and the choice was commercial. Self-hosting and open weights exist in the generator design space — the author's Mowgli stack is a small, deliberately modest existence proof[1] — so the relevant question is not whether self-contained AI is possible but why the dominant presentation is the one where the customer owns least.

For the reader, the practical lesson is unchanged by the political detour, only reinforced: a technology delivered as an opaque service is not merely expensive — it is structurally harder to study, audit, reproduce, and therefore to trust; and, for the dependent customer, harder to keep.

5.3 Misrepresentation and the incentive structure

The incentives that selected the service architecture of Section 5.2 do not stop at shaping how the technology is delivered; they shape what is claimed about it. Vendors are the interested party in their own performance claims, the evaluation instruments most visible to buyers are run by or with those same vendors, and the systems themselves are tuned for human approval. None of this requires dishonesty by any individual to produce systematic misrepresentation — it requires only that each link in the chain be individually reasonable.

The mismatch between marketing and capability is the familiar category error of Table 1 wearing a sales pitch: fluency presented as reasoning, fluent delivery as calibrated knowledge, benchmark gains as understanding. Section 3.6 already documents why headline metrics are unreliable guides; here the point is that the gap between claimed and delivered capability is not an accident of immature evaluation practice but a predictable product of who profits from the claim. A firm competing on perceived capability has every reason to fund demonstrations of success and little to fund quantifications of failure — and the academic freedom to publish the latter has historically been concentrated at exactly the firms with the strongest commercial interest in the former's interpretation.

Two measured findings sharpen the concern beyond marketing rhetoric. The first concerns the products themselves: assistants trained with human feedback consistently exhibit *sycophancy* — agreeing with a user's stated belief, flattering their work, abandoning correct answers under pushback — across every assistant and task family studied, and the behavior traces to human preference data and is amplified when responses are optimized against preference models (Sharma et al. 2023). A system tuned for approval is a system whose apparent agreement is not evidence of correctness; sold to non-experts, it actively interferes with the corrections on which its useful application depends. The second concerns the measurement layer. On the most visible public leaderboard, a small group of large providers was permitted to test many private model variants

[1] *Mowgli* (`https://framagit.org/NadiaYvette/mowgli`) is a small-scale neurosymbolic stack that runs on commodity local hardware with weights, rules, and logic inspectable — an existence proof of the self-contained shape, not a claim of maturity.

in parallel — up to twenty-seven at once — before public release, an asymmetry unavailable to independent researchers; silent deprecation of the majority of public models made rankings unreliably lopsided; and scores proved responsive to surface style — restricted data access during tuning produced relative gains of up to 112% on a downstream benchmark (Singh et al. 2025). The scoreboard that anchors public perception is itself shaped by the firms it ranks.

Overfitting to the tests is the quiet complement. When a fresh set of everday school arithmetic problems was written to match an established benchmark, several model families dropped by up to eight percentage points — systematic overfitting to the original test — though, for honesty's sake, the frontier systems showed little of it (Zhang et al. 2024). Reading that result commercially is instructive: for a firm whose next funding round or product cycle depends on the headline number, teaching to the test is the rational strategy, and nothing in the evaluation market penalizes it. Effort flows to what is measured and sold; comprehension of the kind Section 3.2 found unreliable generates no leaderboard score at all. Whether this slows genuine progress, or merely redistributes it toward the measurable and the saleable, is exactly the kind of open question Section 6 holds apart from the evidence — and the falsification conditions are stated there.

6 Open questions

Distinguishing in-principle limits from in-practice ones is the honest way to hold open questions. The following are open, with the current state of evidence noted.

- **Is confabulation reducible to acceptable levels under an objective that rewards plausibility?** Scaling has diluted but not removed it, and retrieval patches the symptom on popular topics (Mallen et al. 2023) while leaving the long tail exposed. Nothing in the current evidence settles whether the objective itself can be repaired.
- **Do effective and advertised context lengths converge?** The measured gap (Hsieh et al. 2024) and the position pathology (Liu et al. 2024) have persisted across model generations; whether architectural change (including diffusion's all-position access) closes them is unresolved.
- **Is reliable compositionality reachable by training alone?** Errors compound with reasoning depth (Dziri et al. 2023), and the leading current proposal is hybridization rather than pure scale (Kambhampati et al. 2024) — which concedes the in-principle question without proving it.
- **What generally restores lost plasticity?** Shrink-and-perturb and resets work empirically (Dohare et al. 2024; Nikishin et al. 2022); a general theory of when continual learning remains possible is missing.
- **Will diffusion displace autoregression?** The named battlegrounds — reasoning-time compute and long-context economics — are genuine open empirical questions, not settled either way (Section 4).
- **Can the semiotic typing of contextual integrity be formalized?** Filling the theory's informal slot — what kind of sign is transmitted about whom — with Peircean sign classes is, to the author's knowledge, unclaimed formalization work (Section 5.1).
- **Motive, effect, and misrepresentation in the service architecture.** The falsification directions are specific. On motive: release decisions tracking safety review rather than competitive events would weaken the moat account; persistent unrestricted frontier releases would undermine it outright; and regulatory discovery could settle it directly (Section 5.2).

On the political-effect claim: transparency, interoperability, and access metrics recovering while the service model persists would weaken the attribution to architecture. On misrepresentation: if vendor-independent evaluation instruments come to anchor public perception, if leaderboard findings fail replication, or if measured asymmetries (private variant testing, silent deprecation, sycophancy rates) do not persist in deployed systems, the misrepresentation argument of Section 5.3 weakens; their persistence strengthens it.

- **Which limitations survive a well-designed hybrid?** The honest design target is not a system with no limitations but one whose remaining limitations are known and bounded — the question of what that system still cannot do is the one this catalogue cannot yet answer.

7 Conclusion

The generative systems of the past decade are genuinely useful, and nothing in this catalogue should be read as dismissing them. The point of a limitation catalogue is calibration of trust: knowing what a tool cannot do is what makes its successes usable. The catalogue organizes into three families with different causes and different remedies — statistical failures that respond to data and measurement, dynamical failures that respond only to architectural separation, and normative failures that respond only to representable, inspectable structure. Most of the gap between the promises made for these systems and their delivered performance is, on the evidence assembled here, a category error about which family a given problem belongs to.

For the non-specialist reader, the catalogue compresses into a handful of questions worth asking of any claim: Is the evidence observed, explained, or merely narrated? Which family of failure is the claim about? Does the proposed remedy match the family, or is a statistical answer being offered for a dynamical or normative problem? Those questions cut through both hype and dismissal, and they will outlast the current product cycle — because the families, unlike the benchmarks, are not going anywhere.

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A A note on method: verifying a bibliography

Because this document’s central discipline is the separation of the observed from the merely narrated, it would be self-defeating to support that discipline with a bibliography assembled from memory. It was not. Every entry was checked on 2026-09-03 against at least one public bibliographic index — DBLP, the arXiv API, and Crossref — by a small verification script, with the raw machine-readable results and a human-readable corrections log distributed alongside this document.

The verification pass caught real errors worth confessing, because they are representative of how citation folklore forms. Two arXiv identifiers written from memory resolved to entirely unrelated papers (one, remarkably, to a study of intracranial aneurysms); a flagship result’s year and co-author list were wrong in exactly the way that survives years of casual quoting; and one near-collision of titles — there are two different papers beginning “Lost in the Middle” — would have silently redirected a reader. None of these errors would have survived an attempt to download and read the cited papers, which is the deeper point: the failure mode that produces citation folklore is the same one this document catalogues in Section 3.1, running at human speed.

Open-access copies of twenty-nine of the cited papers are archived locally with cryptographic hashes and source URLs recorded, so that the specific evidence behind each claim can be checked rather than trusted. Where a claim rests on a paywalled source, the bibliography says so; where the literature disagrees, the text reports the disagreement rather than resolving it by fiat.